

Vedantu Chemistry Olympiad 2026



VChO-2026

July 19-21, 2026

Instructions

- You are allowed to use Wikipedia or books in this exam. Asking for help on online forums, solving problem with AI or your teachers will be considered cheating.
 - Write your solutions on A4 sheet and make a pdf. Name of the pdf should be same as your team's name. And send the solutions on my telegram ID before 5:00 PM of 21st July.
 - You will get partial marks if some part of question is right.
 - Maximum Marks that you can earn:
 - Problem 1 (12 Marks)
 - Problem 2 (16 Marks)
 - Problem 3 (10 Marks)
 - Problem 4 (22 Marks)
 - Problem 5 (18 Marks)
- Maximum Score: 80 Marks**

Disclaimer: Problems used in this Olympiad are not original but taken from sources all around. So, I want to thank all the creators who have designed such wonderful problems.

Some Important data:

Avogadro number	$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$	$1 \text{ atm} = 101325 \text{ Pa}$
Molar gas constant	$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ $= 0.08205 \text{ L atm K}^{-1} \text{ mol}^{-1}$	Density of water = 1000 kg m^{-3}
Boltzmann constant	$k_B = R/N_A$	

For notes and updates to this table, see www.iupac.org. This version is dated 1 December 2018.
Copyright © 2018 IUPAC, the International Union of Pure and Applied Chemistry.



Problem1: Use of iron in “blue” colours**12 marks**

British astronomer and photographer John Herschel used Fe-based reagents to obtain a blue pigment (known as Turnbull’s Blue) to develop the blue printing technique in 1842. Now, blue printing can be easily carried out using photochemical redox chemistry of potassium ferrioxalate.

Potassium ferrioxalate is synthesized by adding aqueous ferric nitrate to hot aqueous potassium oxalate. To optimize this process, it is important to determine the molar ratio of iron and oxalate in the complex that can be done using redox titrations. In one such analysis, 0.250 g of potassium ferrioxalate sample was taken to a flask and 25 mL of 4M H₂SO₄ was added to it. After making the solution hot, it was titrated with 0.0196 M KMnO₄ till light pink colour was obtained. The observed burette reading was 31.1 mL which indicates oxalate content present in the complex.

- 1.1 Write the balanced half-cell equations for the reaction involved in the titration of oxalate with KMnO₄.

After this titration, excess of zinc powder was added to the same flask that led to effervescence in the solution. When the effervescence subsided, the solution was boiled for 10-15 minutes. The flask was cooled to room temperature and was titrated with the same KMnO₄ solution. (The solution was still acidic). The observed burette reading was 5.1 mL. This will give the iron content in the sample.

- 1.2 Write the possible balanced chemical equation (s) for reaction (s) taking place after Zn dust is added to the flask.
- 1.3 Write the balanced half cell equations for the reaction involved in the titration of iron with KMnO₄.
- 1.4 Calculate the moles of iron and oxalate present in the sample and their molar ratio. (Show the main steps in your calculation).

(i) Calculations for oxalate content:

(ii) Calculations for iron content:

(iii) Molar ratio of iron: oxalate (to the nearest whole number):

The blueprinting utilizes the light-sensitive properties of the trioxalatoferate(III) framework, [Fe(C₂O₄)₃]³⁻. Upon receiving photons in the blue/UV range, an intramolecular charge-transfer process reduces the core iron center. The widely accepted secondary radical decomposition cascade progresses via the following sequential steps:

- **Step 1:** [Fe(C₂O₄)₃]³⁻ + *hν* → [Fe(C₂O₄)₂]²⁻ + C₂O₄^{•-}
- **Step 2:** C₂O₄^{•-} → CO₂ + CO₂^{•-}
- **Step 3:** [Fe(C₂O₄)₃]³⁻ + CO₂^{•-} → [Fe(C₂O₄)₂]²⁻ + CO₂ + C₂O₄²⁻

- 1.5 (i) Write the complete, net balanced ionic equation for the overall light-driven decomposition of the trioxalatoferate(III) complex based on this complete path.
- (ii) The quantum yield (Φ) is defined as the ratio of moles of a specific target product generated to the moles of incoming photons (einsteins) absorbed by the medium. Deduce the precise theoretical value of Φ with respect to the formation of Fe²⁺ ions.

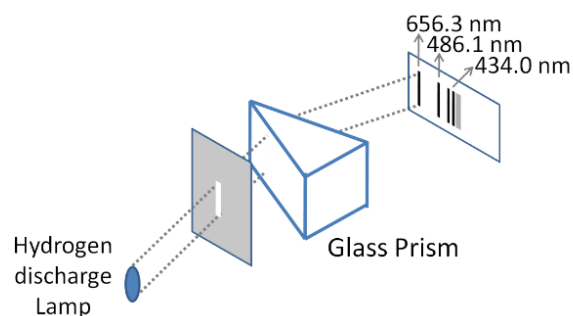
- 1.6 (i) Write the balanced precipitation chemical equation representing the generation of insoluble Turnbull's Blue, structurally localized as the coordination polymer matrix $\text{KFe}[\text{Fe}(\text{CN})_6]$, from the immediate cross-coupling of aquated $\text{Fe}^{2+}(\text{aq})$ and potassium ferricyanide solution, $\text{K}_3[\text{Fe}(\text{CN})_6](\text{aq})$.
- (ii) Assume the total iron mass quantified via the 5.1 mL titration step in question 2.17 is isolated and completely converted into dry, anhydrous $\text{KFe}[\text{Fe}(\text{CN})_6]$ solid. Calculate the maximum theoretical absolute mass (in mg) of the blue pigment that could be harvested from this operation. (Molar Mass of $\text{KFe}[\text{Fe}(\text{CN})_6] = 306.89 \text{ g}\cdot\text{mol}^{-1}$).

Problem 2

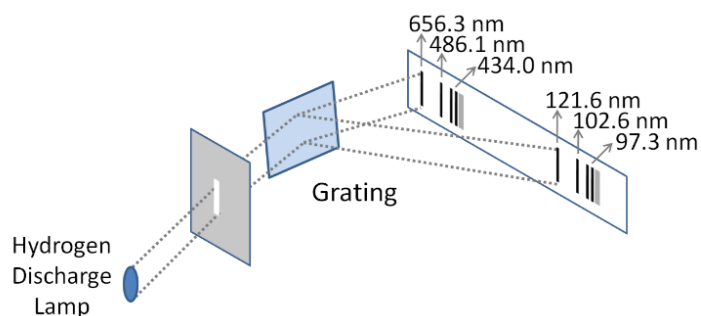
16 marks

The purple hydrogen

When high voltage is applied to hydrogen gas enclosed in an electric discharge tube, it gives a purple glow. In nature, such colour can be observed in light coming from some stars. Origin of this colour in hydrogen remained a mystery for long. In an experiment in the 19th century, scientists passed the light from a hydrogen discharge tube through a slit followed by a glass prism. On a screen kept behind the glass prism, multiple lines could be seen with a very bright red line followed by a blue and few closely spaced violet lines. In experiments in early 20th century when diffraction grating was used instead of glass prism, few additional lines could be detected as non-visible extensions of the spectrum as shown below.



Spectrum S1



Spectrum S2

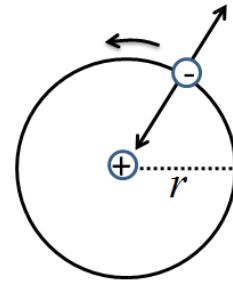
In later experiments, several more lines at higher wavelengths were detected in the spectrum of hydrogen discharge, but no lines were detected at wavelengths lower than those seen in *spectrum S2*.

- 2.1 Each of these lines originated from a specific decrease in energy of hydrogen atoms as the atoms transitioned from one well-defined energy state to another. Determine the frequency of light originating from the lowest energy transition observed in *spectrum S2*.

The wavelengths obtained in the above spectra were analyzed by many mathematicians. Finally, it was recognized that the wavelengths corresponded to change in energy states of hydrogen atom, where energy of each state would be of the form $E_n = -C/n^2$, where C is a constant and n is a natural number.

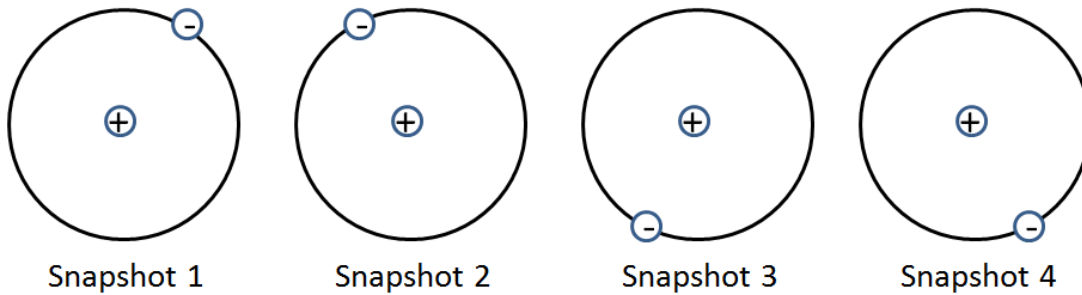
- 2.2 i) Using the wavelength data in *S1* and *S2*, determine the value of C in Joules and in cm^{-1} . Give detailed steps followed to obtain the answers.
- ii) Determine the n values for states, transitions between which produces light of wavelengths 102.6 nm and 434.0 nm.

According to the Bohr atomic model, electrons in an atom move in circular orbits around the nucleus (as depicted in the figure). The hydrogen atom is stable because the magnitude of the centrifugal force $\left(\frac{mv^2}{r}\right)$ and electrostatic force $\left(\frac{Ze^2}{4\pi\epsilon_0 r^2}\right)$ at distance, a_0 are equal.

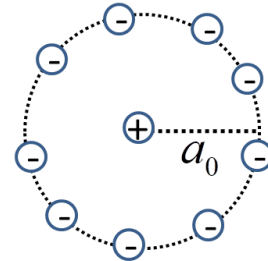


2.3 Find the magnitude of velocity of the electron in hydrogen atom and find its ratio to the speed of light.

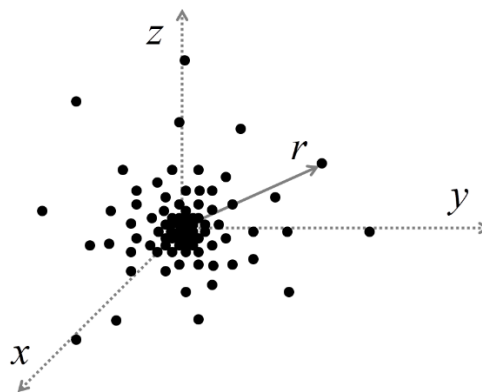
Imagine that we can capture snapshots of the electron orbiting in the circular path. Four representative snapshots among several are illustrated below:



Alternatively, several snapshots can be collectively represented in a single figure as given on the right-hand side image. This figure shows that the probability of finding the electron on the orbit of radius $r = a_0$ is 100% (according to the Bohr model).



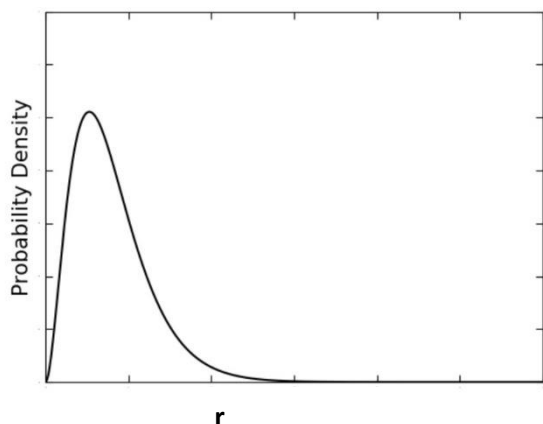
On the contrary, different experiments collectively exhibit the following superimposed snapshots for the position of the electron in a hydrogen atom. Here, the nucleus is located at the origin and position of the electron (shown by dots) can be anywhere around the nucleus; however, there are regions where the electron is found more frequently than the other regions.



Quantum mechanical model provides a possible explanation for the above experimental realization using a mathematical function, called the *probability density function*. The probability density of the electron in hydrogen atom or hydrogen like atoms/ions (with atomic number Z) in lowest energy state at a distance r obtained from quantum mechanical model, can be expressed as:

$$f(r) = \frac{4Z^3}{a_0^3} r^2 e^{-2\frac{Z}{a_0}r}$$

This function for hydrogen is plotted in the following figure:

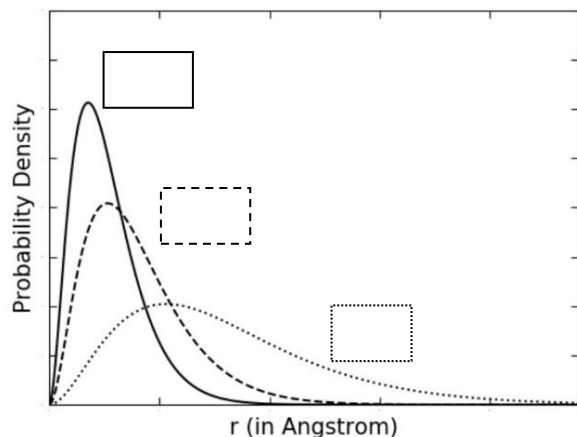


2.4 Find the most probable value of r (the distance from the origin where the probability density of finding the electron is maximum) for a hydrogen atom.

2.5 Based on the most probable value of r , possible true statements among the following is/are (Mark **X** for the correct option(s)):

- a) Bohr model predicted locations of electron which are impossible in reality.
- b) In quantum mechanical model, circular motion is not possible, only linear motion is possible.
- c) In quantum mechanical model, electron motion is not possible.
- d) In quantum mechanical model, electron in lowest energy state can be found at a distance $r = 8a_0$.

2.6 Curves representing the ground state probability density functions of electron in H atom, He^+ ion and Li^{2+} ion are given in diagram below. Identify which curve corresponds to which specie.



Assume that a specially designed hydrogen discharge lamp **DL** creates excited hydrogen atoms in $n = 2$ state only. Electronically excited atoms finally transition back to the ground state by emitting photons with appropriate frequency. One can monitor the emission intensity, I at any given time, t , using a photosensitive detector.

Assume that at time $t = 0$ (i.e., when the excited hydrogen atoms were generated), the emission intensity was I_0 . Change of emission intensity as a function of time can be expressed by the following rate equation:

$$-\frac{dI}{dt} = kI ; \text{ where, } k = 10^8 \text{ s}^{-1} \text{ for } n = 2 \rightarrow n = 1 \text{ transition and depends on the electron distribution}$$

in the initial and final states.

2.7 Find the time at which the intensity of discharge lamp drops to 10% of the initial intensity after it is switched off.

2.8 Identify the colour of emission of this discharge lamp **DL**.

Problem 3: Absorption of Light by Molecules**12 marks**

Absorption of light by molecules is the first step of all photochemical reactions. The Beer - Lambert law relates the absorbance A of a solution containing an absorbing species of molar concentration c with the optical path length d when incident power is P_0 and transmitted power is P :

$$A = \log \frac{P_0}{P} = \epsilon c d \quad \epsilon \text{ is the molar absorptivity (also called extinction coefficient).}$$

Light can be considered as a stream of photons, each carrying an energy of $E = h \frac{c}{\lambda}$;

h is Planck's constant, λ is the wavelength and c the speed of light .

A solution with a dye concentration of $c = 4 \cdot 10^{-6} \text{ mol L}^{-1}$ has a molar absorptivity of $\epsilon = 1.5 \cdot 10^5 \text{ mol}^{-1} \text{ L cm}^{-1}$. It is illuminated with green laser light at a wavelength of 514.5 nm and with a power of $P_0 = 10 \text{ nW}$.

3.1 What is the percentage of light that is absorbed by the sample after a path length of 1 μm ?

3.2 Calculate the number of photons per second absorbed by the sample.

The absorption cross section of a molecule is the effective area that captures all incoming photons under low illumination conditions (like an idealized solar cell that would capture all light photons hitting its surface). At room temperature, this corresponds roughly to the molecular area exposed to the light beam. If you calculate it from the molar absorptivity, imagine that all molecules interacting with the light are arranged periodically in a plane perpendicular to the incoming light beam.

3.3 What area is occupied by each molecule?

3.4 Calculate the molecular absorption cross section in units of $\text{\AA}^2$.

A crucial photochemical reaction for life on our planet is photosynthesis, which converts the absorbed light energy into chemical energy. One photon of 680 nm is necessary to produce one molecule of ATP. Under physiological conditions, the reaction requires an energy of 59 kJ per mol of ATP.

3.5 What is the energy efficiency of photosynthesis?

Part A: Particle in a one-dimensional box

The simplest Schrödinger equation, describing a free particle confined to move in a one-dimensional 'rigid box' brings out a most basic fact: quantization arises due to boundary conditions on the wave function.

An electron of mass m is confined to move in a line along the x -axis from $x = 0$ to $x = L$. Between the two ends it experiences no force.

4.1 Write down the (time-independent) Schrödinger equation for the wave function ψ of an electron.

4.2 Which of the following are possible wave functions of an electron in one-dimensional rigid box :

$$\begin{aligned} &e^{-kx} \\ &\cos \frac{n\pi x}{L} \\ &\sin kx \\ &\sin \frac{n\pi x}{L} \end{aligned}$$

where k is any real number and n is a positive integer ?

4.3 For the acceptable wave functions of the electron in (ii) above, show that the energies are given by

$$E_n = \frac{h^2 n^2}{8mL^2}$$

4.4 Plot schematically the wave function of the electron in the ground and the first two excited states. What is the number of nodes (in the region between $x = 0$ to L) of the wave function with energy E_n ?

4.5 Normalize the ground state wave function of the electron.

(The integral of the square of the modulus of a normalized wave function over all space is unity.)

Part B: A Particle in a 2-D Box

The π electrons of the iron-heme of a hemoglobin molecule can be visualized as a system of free electrons moving in a two-dimensional box. According to this model, the energy of the electron is limited to the values

$$E_{n_x, n_y} = \frac{h^2}{8m_e L^2} (n_x^2 + n_y^2), \quad n_x, n_y = 1, 2, 3, \dots$$

where $h = 6.63 \times 10^{-34}$ J s is the Planck constant; n_x and n_y are the principal quantum numbers; $m_e = 9.11 \times 10^{-31}$ kg is the electron mass; L is the length of the box.

- 4.6 Construct an energy level diagram showing the relative ordering of the lowest 17 orbitals.
- 4.7 Given the molecule contains 26 electrons, determine the electron population of the highest occupied orbitals in the ground state.
- 4.8 Assuming Hund's rule can be applied to this system, predict whether or not this system is paramagnetic.
- 4.9 Light is absorbed only when the condition $h\nu = \Delta E$ is satisfied. If the length L for this 2D box is 1 nm, what is the longest wavelength of light that can lead to excitation? Express your result in nm. [The speed of light, $c = 3.00 \times 10^8$ m/s]

Problem 5: Laser Cooling

18 Marks

This question is about laser cooling, which is a quick and efficient way of cooling ions down to very cold temperatures. The mean kinetic energy of a molecule is related to its temperature by $E = \frac{3}{2} k_B T$, where k_B is the Boltzmann constant.

- 5.1 Calcium atoms leak out of an oven at 600 °C. Calculate the mean kinetic energy of the calcium atoms and hence the rms momentum and rms speed of a ^{40}Ca atom, whose relative isotopic mass is 39.96.
- 5.2 The atoms drift into an ion trap where they are photoionised and trapped. While in this trap they are bombarded with laser light of wavelength 396.96 nm. Calculate the frequency, energy and momentum of a photon with this wavelength.

- 5.3** The ions go through an optical cycle repeatedly. Ions absorb a photon from the laser when they are moving in the opposite direction to the light (this is achieved using the Doppler Effect) and then re-emit a photon in a random direction. The net effect of this procedure is to slow the ion down slightly. Calculate the change in mean momentum and speed at each cycle and the number of photons that would need to be absorbed to bring the ion approximately to rest. (In practice this process was found to reduce the temperature to about 0.5 mK.)
- 5.4** Write down the ground electronic configuration of the Ca^+ ion, and calculate the orbital and spin angular momentum of the unpaired electron.
- 5.5** In the excited configuration involved in the laser cooling transition the unpaired electron has been excited into the lowest available p orbital. Calculate the orbital and spin angular momentum of the unpaired electron.
- 5.6** In this excited state the electron experiences a magnetic field because of its own orbital motion around the charged nucleus. The spin of the electron can line up either parallel or antiparallel to this field, and the two states have slightly different energies. The resultant quantum number, j , for the total electronic angular momentum takes values from $|l - s|$ to $|l + s|$ in integer steps. Calculate the possible values of j .
- 5.7** The laser cooling transition is to the lower of these two levels, the transition from the ground state to the higher level has a wavelength 393.48 nm. Calculate the energy difference between the two levels resulting from the excited configuration.

End of the paper